

WHITE PAPER

Mandum Rimba

One open map for Indonesia's forests.

Bringing together public data on Indonesia's forests, wildlife, and disasters into one map that is easy to read. We summarize and display the data as it is, without judging or drawing conclusions.

An independent, nonprofit open data map

Version 1.1

Open source (MIT license)

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Summary

Each year Indonesia's forests grow smaller, and animals such as elephants, tigers, and orangutans lose the space they need to live.

The data about these forests is already public. Satellite imagery, area boundaries, and disaster records are all available. The trouble is that this data is scattered across many technical reports that ordinary people rarely open, so it is hard to see the full picture.

Mandum Rimba gathers that public data into one map that is easy to read, free, and open to anyone. We do not analyse, judge, or draw conclusions. We simply summarise and display the data as it is, together with its source, and leave the interpretation to the reader.

In one sentence

Mandum Rimba is an open map that brings public data on Indonesia's forests and wildlife into one place, so that anyone can see and understand it more easily.

1 Why this map exists

News about changing forests and animals losing their habitat often passes by very quickly. What we do not see is easy to forget.

Yet the data exists. Satellites map forest cover almost every day, and many institutions make their data public. The problem is not a lack of data, but its form, which is scattered, technical, and hard for ordinary people to reach.

Mandum Rimba began with a simple wish: to make data that is already open genuinely visible and understandable to many people, in friendly language and a single, complete view.

2 What Mandum Rimba is

The name Mandum Rimba comes from the Acehnese language. Mandum means all, and Rimba means wilderness, the home of our wildlife. Together, Mandum Rimba means the forest that belongs to us all, not to any single party or generation, but a shared home for the people and animals that live within it.

The name deliberately uses a local language, because this concern began in Aceh, with forests that looked thinner year after year. That is where our data first started.

Where we stand

Mandum Rimba is an independent, nonprofit open data map. It is a contribution to the public, not a commercial venture. We do not sell data, we do not run advertising, and we do not represent the interests of any party.

Four principles we hold to

- **Independent.** Not tied to any commercial or political interest.
- **Nonprofit.** No data is sold and no advertising is shown.
- **Open source.** All of the code and the way data is processed can be inspected by anyone.
- **Neutral.** We present the data as it is, without judging and without drawing conclusions.

3 Why it matters

The state of the forest touches many things at once, from wildlife to people to those who make a living from the land. Presenting the data clearly helps us see the bigger picture.

For wildlife

Forests are home to Sumatran elephants, tigers, and orangutans. As forests shrink, the space these animals have shrinks with them. Showing this on a map helps us see something that often escapes attention.

For people

The state of the forest is closely connected to the lives of many people, from clean water to disaster preparedness. Data that is easy to see helps us understand the situation better.

For those who live off the land

Many people depend on the land for their livelihood, and we are not here to judge anyone. We simply display the data so the picture is more complete for everyone.

4 How we work

Every data layer in Mandum Rimba has a source, a date of collection, and a clear description. This section explains where the data comes from and how we present it.

Data sources

We do not create data ourselves. We gather data that has already been opened by trusted institutions and scientists, and bring it together in one view. The main sources are below.

Source	What it provides	License
Global Forest Watch (WRI)	Daily forest cover change alerts (RADD and GLAD) and annual tree cover loss	CC BY 4.0
GFW (concession layers)	Concession boundaries: oil palm, wood fibre, and timber	CC BY 4.0

Source	What it provides	License
Maus et al. 2022 (PANGAEA)	Global mining land footprint	CC BY 4.0
BNPB DIBI / UNDRR	Records of flood and landslide events	Public data
GBIF	Occurrence records of threatened species	Per dataset
RESOLVE Ecoregions	Ecoregions as wildlife habitat units	CC BY 4.0
Protected Planet (WDPA)	Protected area boundaries	WDPA terms
KLHK PIPPIB	Forest moratorium areas	Public data
Trase	Commodity supply chain data	CC BY 4.0
GADM 4.1	Administrative boundaries	Academic
HydroBASINS	River basin (watershed) boundaries	HydroSHEDS

Bringing together, not concluding

We place each data source as a separate layer on the same map. Readers can turn each layer on or off and see the picture for themselves.

We do not combine data to draw conclusions, we do not assign judgements, and we do not point at any party. Every figure is shown as it is and can be traced back to its original source.

Summarising, not concluding

Mandum Rimba does not judge and does not accuse. Its only purpose is to summarise public data so it is easier to see. Any interpretation is left entirely to the reader.

5 How we keep trust

A map is only useful if people can check it, rather than simply believe it. That is why we built Mandum Rimba so that every figure can be traced.

Every figure is traceable

Everything shown on the map can be traced back to its source dataset, along with its date of collection.

It can be checked

The way we gather and display data is open, so it can be inspected and rerun by anyone.

Recorded and transparent

Every data update is logged openly, so changes can always be seen over time.

6 Limitations worth knowing

We believe that being honest about limitations actually strengthens trust. Here are some things to keep in mind when reading this map.

- The data comes from various sources with different update dates, so it does not always reflect the most current conditions.
- Satellite imagery has limits to its precision, and small objects can be missed.
- Area and boundary data follow the public data available, which may be updated by its source at any time.
- Because we only display data, this map is best read as a starting point for further exploration, not as a final conclusion.

We note the source and date on every layer, so that everything can be traced back easily.

7 The technology behind it

Mandum Rimba is built entirely on open source tools, so anyone can inspect it or rebuild it.

- The map is drawn with MapLibre GL JS, using static PMTiles vector tiles built with tippecanoe.
- Geospatial data is stored and managed with MongoDB.
- The web side uses Next.js, while data services run on NestJS.
- The site is available in two languages, Indonesian and English, through next-intl.

8 An invitation to collaborate

Mandum Rimba is built slowly as a way of giving back to the open source and open data communities, where so much was learned for free. This project will be far better if it is built together.

With all humility, we invite friends at environmental organisations, data communities, academics, and anyone who cares to take part.

What would help

Additional data, feedback on how things are presented, local knowledge about an area, or help in checking accuracy. However small the contribution, we are deeply grateful.

All of the code is open source under the MIT license, and contributions are open to anyone.

How to reach us

Share your input through the Threads account @r.rasyidi, or through the publicly available code repository.

9 License and credits

The Mandum Rimba source code is licensed under MIT. Each dataset remains the property of its own source, under its own license. Full details are available on the Data Sources and Credits pages.

Thank you to Global Forest Watch (WRI), GBIF, BNPB DIBI and UNDRR, RESOLVE Ecoregions and UNEP-WCMC, Maus et al. 2022, Protected Planet (WDPA), KLHK, GADM, HydroBASINS, and to all the open source tools that keep this project running.

Note

Everything on this map and in this document is presented descriptively from open public data. Mandum Rimba does not judge, does not draw conclusions, and is not an accusation against any party.